

Team Worksheet — Lab 22

Solution Generation, Selection and Lean Countermeasures — In Depth

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-22-solution-selection.xlsx`.

3. What we produced

Deliverable: A scored solution selection matrix mapping each solution to a proven root cause.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Take your ranked proven root causes from Lab 21. Every solution generated in this lab must trace to one of them.

Step 2. Recap divergent generation techniques in full: brainstorming, brainwriting (6-3-5), SCAMPER, anti-solution thinking and analogy from other industries.

Step 3. Recap benchmarking and its four steps — plan, collect, analyse, adapt. Look outside your industry, where the best analogous processes usually are.

Step 4. Generate at least three candidate solutions for each proven root cause. One solution per cause means you have not really explored the space.

Step 5. Recap the Lean countermeasure toolkit in full: 5S, poka-yoke, pull systems, JIT, Heijunka (levelling), Jidoka (autonomation) and standard work.

Step 6. Recap poka-yoke levels specifically: prevention (error impossible), detection (error caught immediately) and warning (error flagged). Always prefer prevention.

Step 7. Map each Lean tool to the root causes in your capstone it could plausibly address, and generate at least two poka-yoke ideas.

Step 8. Build the weighted solution selection matrix. Criteria typically: impact on Y, implementation cost, time to benefit, technical risk, ease of adoption and sustainability.

Step 9. Weight the criteria with your sponsor, not alone — the weighting reflects business priorities that are theirs to set.

Step 10. Score every candidate solution and compute weighted totals.

Step 11. Sanity-check the top-ranked solutions: do they collectively address the causes explaining most of the Y gap? If not, revisit generation.

Step 12. Identify which solutions have controllable factors suitable for DOE optimisation in Labs 23-25.

7. Self-check

Every solution traces to a proven root cause, criteria weights are agreed with the sponsor, and your selected set collectively addresses the majority of the quantified Y gap.

Complete? ■ Yes ■ Not yet

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